Siru Ouyang

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Education

University of Illinois Urbana-Champaign — **Urbana, USA**
Aug. 2022 - present

- Ph.D. in Computer Science
- Advisor: Professor Jiawei Han

Shanghai Jiao Tong University — **Shanghai, China**
Sep. 2018 - Jun. 2022

- B.Eng. in Computer Science (IEEE honor class)
- Advisor: Professor Hai Zhao, Professor Xiaofeng Gao

Research Interests

- Self-Evolving LLM Agents, Agentic Post-Training, Complex Reasoning in LLMs

Selected Publications

For the full list, please refer to the Google Scholar Page.

SkillOS: Learning Skill Curation for Self-Evolving Agents with Reinforcement Learning

- Siru Ouyang, Jun Yan, Rui Meng, Bhavana Dalvi Mishra, Chun-Liang Li, Zifeng Wang, Rujun Han, Yizhu Jiao, Maohao Shen, Kaiwen Zha, George Lee, Hangfei Lin, Jiawei Han, Chen-Yu Lee, Tomas Pfister
- In submission to **NeurIPS 2026**

MemoryArena: Benchmarking Agent Memory in Interdependent Multi-Session Agentic Tasks

- Zexue He, Yu Wang, Churan Zhi, Yuanzhe Hu, Tzu-Ping Chen, Lang Yin, Ze Chen, Tong Arthur Wu, Siru Ouyang, Zihan Wang, Jiaxin Pei, Julian McAuley, Yejin Choi, Alex Pentland
- **ICML 2026**

ReasoningBank: Scaling Agent Self-Evolving with Reasoning Memory

- Siru Ouyang, Jun Yan, I-Hung Hsu, Yanfei Chen, Ke Jiang, Zifeng Wang, Rujun Han, Long Le, Samira Daruki, Xiangru Tang, Vishy Tirumalashetty, George Lee, Mahsan Rofouei, Hangfei Lin, Jiawei Han, Chen-Yu Lee, Tomas Pfister
- **ICLR 2026**, featured by VentureBeat, MarkTechPost, and Google Blogpost

Discourse-Aware Language Representation

- Zhuosheng Zhang, Siru Ouyang, Hai Zhao
- **IEEE Transactions on Pattern Analysis and Machine Intelligence (Impact Factor: 18.6)**

RAST: Reasoning Activation through Small-model Transfer

- Siru Ouyang, Xinyu Zhu, Zilin Xiao, Minhao Jiang, Yu Meng, Jiawei Han
- **NeurIPS 2025**

RepoGraph: Enhancing AI Software Engineering with Repository-level Code Graph

- Siru Ouyang, Wenhao Yu, Kaixin Ma, Zhihan Zhang, Zilin Xiao, Mengzhao Jia, Jiawei Han, Hongming Zhang, Dong Yu
- **ICLR 2025**

ChemAgent: Self-updating Memories in Large Language Models Improves Chemical Reasoning

- Xiangru Tang, Tianyu Hu, Muyang Ye, Yanjun Shao, Xunjian Yin, Siru Ouyang, Wangchunshu Zhou, Pan Lu, Zhuosheng Zhang, Yilun Zhao, Arman Cohan, Mark Gerstein
- **ICLR 2025**

Structured Chemistry Reasoning with Large Language Models

- Siru Ouyang, Zhuosheng Zhang, Bing Yan, Xuan Liu, Yejin Choi, Jiawei Han, and Lianhui Qin.
- **ICML 2024**

Does Data Contamination Make a Difference?
Insights from Intentionally Contaminating Pre-training Data for Language Models

- Minhao Jiang, Ken Ziyu Liu, Ming Zhong, Rylan Schaeffer, Siru Ouyang, Jiawei Han, Sanmi Koyejo
- **Best paper award** at **DPFM Workshop ICLR 2024**.

The Shifted and The Overlooked: A Task-oriented Investigation of User-GPT Interactions

- Siru Ouyang, Shuohang Wang, Yang Liu, Ming Zhong, Yizhu Jiao, Dan Iter, Reid Pryzant, Chenguang Zhu, Heng Ji, and Jiawei Han.
- **EMNLP 2023**

Research Experiences

Google Cloud AI Research — **Sunnyvale, CA**
May 2025 - May 2026

- Student Researcher
- Host: Dr. Jun Yan
- Research Topic: Scaling Agent Self-Evolving with Reasoning Memory; RL for skill management

Tencent AI Seattle Lab — **Seattle, WA**
May 2024 - Aug. 2024

- Research Intern
- Host: Dr. Wenhao Yu, Dr. Kaixin Ma
- Research Topic: Towards repository-level understanding with plug-and-play code graph

Microsoft Research, Redmond — **Remote**
May 2023 - Dec. 2023

- Research Intern, Cognitive Service Research Group and Azure AI
- Host: Dr. Shuohang Wang
- Research Topic: User-centric investigation of LLMs and NLP efficiency

Georgia Institute of Technology — **Remote**
June 2021 - Dec. 2021

- Research Intern, School of Interactive Computing
- Advisor: Prof. Diyi Yang
- Research Topic: Data augmentation for abstractive conversation summarization

Honors and Awards

- **NeurIPS Scholar Award**, Neural Information Processing Foundation, 2025.
- **WWW Student Travel Grant Award**, The Web Conference. 2025
- **Richard T. Cheng Endowed Fellowship**, University of Illinois Urbana-Champaign. 2022
- **Outstanding Graduate at SJTU**, Shanghai Jiao Tong University. 2022
- **Rongchang Scientific and Technological Innovation Scholarship**(Rank **1**st in the School of Electronic Information and Electrical Engineering), Shanghai Jiao Tong University. 2021
- **SenseTime Scholarship** (31 out of all students for artificial intelligence in China), SenseTime Incorporation. 2021
- **Google Women Tech Makers Scholarship** (110 out of 2800 students around the whole Pacific Rim), Google. 2020
- **SJTU Academic Scholarship** (5% of all the students in the School of Electronic Information and Electrical Engineering), Shanghai Jiao Tong University. 2019 and 2020